

01_FOG_ORCHESTRATOR_SYSTEM_OVERVIEW

Document ID: DOC-01-OVERVIEW | Date: 2026-09-28 | Audited Standard: SIH Evaluator Evidence

EVIDENCE PROVENANCE: [SYSTEM SYNTHESIS]

Executive overview integrating problem definition, physical prototype implementation, digital twin architecture, and operational differentiators.

1. Operational Problem & Industry Gap

Open-cast mining at NMDC Bailadila (Deposit 5) experiences seasonal valley fog reducing visibility below 15m. A loaded 165.5-tonne BEML BH100 haul dumper on an 8-12% downhill ramp requires 35 to 50m stopping distance. Current industry practice is either blind driving (fatal collision hazard) or blanket shutdown (costing millions of INR/hr). FOG-ORCHESTRATOR 2.0 provides continuous physics-constrained speed guidance, allowing safe operation down to crawl speeds without shutdown.

2. The FOG-ORCHESTRATOR Solution

- Weather Ingestion:** Haul road optical visibility and road friction continuously measured and ingested.
- Authoritative 3D Twin:** Single state store mirrors mine topography, ramp grades, and dynamic vehicle states.
- 5-Constraint Safety Governor:** Computes maximum defensible safe speed based on stopping distance, friction, retarder limits, curves, and mine rules.
- Driver Assistance:** In-cab cockpit console gives visual and audible speed guidance. Driver remains in full control.
- Dual-Link RF:** 433 MHz LoRa and Wi-Fi dual-link telemetry with automatic emergency Safe Beacon failover.

3. Core Technical Differentiators

Feature	Conventional Practice	FOG-ORCHESTRATOR 2.0
Low-Visibility Pacing	Blanket fleet shutdown (0 t/h)	Regulated safe crawl retaining 68.4% throughput
Safety Enforcement	Driver visual estimation	Inviolable 5-constraint real-time safe speed ceiling
Vehicle Authority	Autonomous vs human confusion	Driver remains in full control; system advises
Digital Twin Role	Static visual dashboard	Authoritative cyber-physical state store
Communication Resilience	Single cellular/Wi-Fi link	Dual-link LoRa (433 MHz) + Wi-Fi + CAN failover

4. Subsystem Integration Summary

Physical prototype features dual ESP32 vehicles with optical wheel tachometers, MPU6050 IMU, L298N and TB6612FNG H-bridges, and SX1278 LoRa transceivers. Software suite provides FastAPI telemetry ingestion, authoritative state store, Control Room 3D console (Port 5173), and In-Cab Driver Screens (Ports 3001/3002).